

ES111 CALCULUS

Promotion:

First Name:

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Question 1: Limits

Discuss the following topics:

- Intuitive Notion of Limit
- Formal Definitions of Limit
- Proof of Unicity
- Rules and Theorems
- Proof of Limit of a Sum
- Proof of Squeeze Theorem
- Limits and Infinity
- Link with Completeness of Real Number
- Proof the Archimedean principle

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Question 2: Continuity and Intermediate Value Theorem

Discuss the following topics:

- Definitions of Continuity
- Continuous Functions and Extensions
- Intermediate Value Theorem
- Proof of Arzelà-Ascoli Theorem
- Proof of Cauchy Theorem
- Proof of Bolzano's Theorem
- Proof of Intermediate Value Theorem
- Application of Intermediate Value Theorem
- Link between Nested Intervals Theorem and Completeness of Real Numbers

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Question 3: Continuity and Extreme Value Theorem

Discuss the following topics:

- Definitions of Continuity
- Continuous Functions and Extensions
- Extreme Value Theorem
- Proof of (Strong) Capture Theorem
- Extreme Value Theorem
- Proof of Boundedness Theorem
- Proof of Extreme Value Theorem
- Link between Theorem of Hein-Borel and Completeness of Real Numbers

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Question 4: Derivatives

Discuss the following topics:

- Intuitive Notion of Derivatives (Tangent Lines)
- Definitions of Derivative
- Differential
- Differentiation Rules
- Proof that a differentiable function is continuous
- Proof of Product Rule, Reciprocal Rule and Chain Rule
- Implicit Differentiation
- Definition of Inverse Function
- Derivative of Inverse Functions

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Question 5: Derivatives, Mean Value Theorem and Indeterminate Forms

Discuss the following topics:

- Definition of Derivative
- Mean Value Theorem
- Proof of the link between a continuous function achieving a maximum in an open interval and the critical value
- Proof of Rolle's Theorem
- Proof of Mean Value Theorem
- Proof of the Generalized Mean Value Theorem
- L'Hôpital Rules
- Proof of the First l'Hôpital Rule

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Question 6: Derivatives, Mean Value Theorem and Extreme Values

Discuss the following topics:

- Definition of Derivative
- Mean Value Theorem
- Proof of the link between a continuous function achieving a maximum in an open interval and the critical value
- Proof of Rolle's Theorem
- Proof of Mean Value Theorem
- Proofs about Increasing and Decreasing Functions
- Critical Points, Singular Points and Endpoints
- First Derivative Test
- Convexity/Concavity and Inflections
- Second Derivative Test

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Question 7: Derivatives, Mean Value Theorem and Taylor Polynomials

Discuss the following topics:

- Definition of Derivative
- Mean Value Theorem
- Proof of the link between a continuous function achieving a maximum in an open interval and the critical value
- Proof of Rolle's Theorem
- Proof of Mean Value Theorem
- Linear Approximations
- Proof of Error Term of a Linear Approximation
- Taylor Theorem
- Proof of Taylor Theorem

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Question 8: Power, Polynomial and Rational Functions

Discuss the following topics:

- Definition of Power Function
- Definition of Derivative
- Complete Proof of $(x^r)' = rx^{r-1}$ ($r \in \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$)
- Definition of Polynomial Functions
- Proof of Continuity of a Polynomial Function
- Derivatives of Polynomial Functions
- Definition of Rational Functions
- Proof the Quotient Rule of Derivation
- Derivatives of Rational Functions

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Question 9: Exponential and Logarithmic Functions

Discuss the following topics:

- Definition of Derivative
- Antiderivative
- Definition of Natural Logarithm
- Properties of Natural Logarithms
- Proof of $\ln\left(\frac{x}{y}\right) = \ln x - \ln y$
- Exponential Function
- Properties of Exponential Functions
- Proof of $(\exp x)^r = \exp(xr)$
- Constant of Euler
- General Exponentials and Logarithms
- Logarithmic Differentiation

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Question 10: Trigonometric Functions

Discuss the following topics:

- Definition of Trigonometric Functions
- Proof of Continuity of Sine
- Proof of $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$
- Definition of Derivative
- Derivatives of Trigonometric Functions
- Definition of Inverse Function
- Derivative of Inverse Functions
- Inverse Sine
- Other Inverse Trigonometric Functions
- Hyperbolic Functions

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Question 11: Integration

Discuss the following topics:

- Intuitive Notion of Integration
- Definition of Definite Integral
- Proof of Theorems about Partitions
- Link between Darboux and Riemann Integral
- Proof of Darboux's Integrability Theorem
- Proof of the Integrability of a Continuous Functions
- Link between Theorem of Hein-Borel and Completeness of Real Numbers

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Question 12: Integration and Mean Value Theorem

Discuss the following topics:

- Definition of Definite Integral
- Darboux's Integrability Theorem
- Proof of the Linearity of the Definite Integral
- Proof of the Mean-Value Theorem for Integrals
- Integral of Piecewise Continuous Functions

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Question 13: Integration and Fundamental Theorem of Calculus

Discuss the following topics:

- Definition of Definite Integral
- Darboux's Integrability Theorem
- Fundamental Theorem of Calculus: Continuity, Derivative and Integral
- Proof of Fundamental Theorem Part 1
- Proof of Fundamental Theorem Part 2
- Generalized Chain Rule

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Question 14: Integration, Volumes by Slicing and Arc Length

Discuss the following topics:

- Definition of Definite Integral
- Link between Darboux and Riemann Integral
- Volumes by Slicing
- Solids of Revolution: rotation about x -axis and y -axis
- Polygonal Approximation of Arc
- Proof the Formula of the Arc Length of the Graph of a Function
- Areas of Surfaces of Revolution: rotation about x -axis and y -axis

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Question 15: Parametric and Polar Curves

Discuss the following topics:

- Definition of (Smooth) Parametric Curves
- Proof the formula for the Slope of the Tangent Line
- Concavity of a Parametric Curve
- Arc Length and Area of Parametric Curves
- Definition of Polar Curves
- Area and Arc Length of Polar Curves
- Importance of Continuity in the Definition of Parametric Curves